

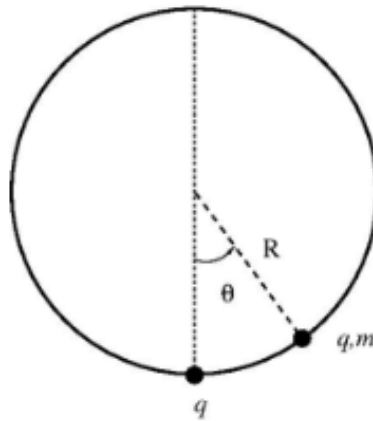
10th Bangladesh Physics Olympiad 2020

National Round Exam

Category-C

Rolling on a Hoop

A point mass of mass m and charge q moves (obviously without rolling!) along a circle of radius R in a vertical plane, under the action of gravity. Now a second charge q is fixed to the bottom most point of the circle (See the figure).



(a) Find the potential energy of the mass as a function of the angle θ .

[1]

(b) Find the equilibrium mass in terms of the other fixed variables.

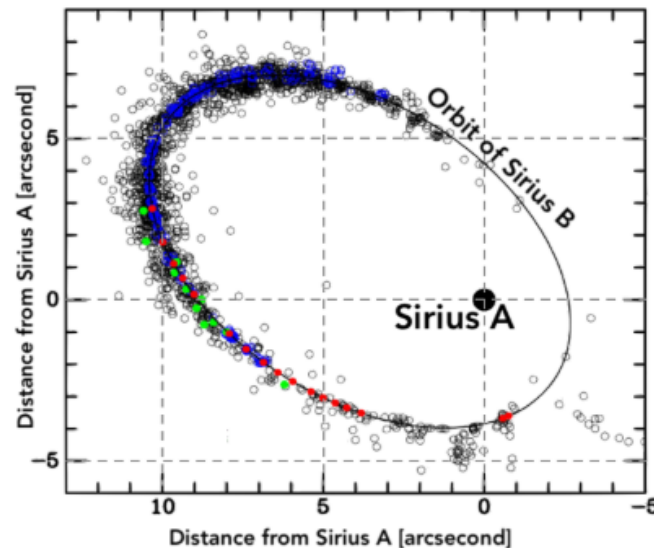
[2]

(c) Find the oscillation frequency of the mass around this equilibrium position.

[2]

Looking at the Twins

The mirror of Hubble Space Telescope (HST) has a diameter of 2.4 m. The brightest star in our sky, Sirius, is actually a binary star system where two stars, Sirius A and Sirius B, are orbiting around each other. The figure shows the orbit of Sirius B around Sirius A. We could not distinguish between the two stars with naked eye even if the second star was much bigger and visible in the night-sky. Note that the "distance" is given in angular units.



- (a) Can HST actually distinguish between Sirius A and B? To put it another way: if you take a picture of the Sirius stars with HST, will you be able to see the two stars in the picture, or will they appear to be merged in one? Give details in support of your answer. [Hints: The resolution of a telescope (the minimum distance it can distinguish) depends on diffraction. Telescopes are like single-slits and light passing through them or reflected from them go through Fraunhofer diffraction. Their resolution is almost equal to the width of the central bright fringe of the diffraction pattern. Use the average wavelength of visible light (550 nanometers) as the wavelength in your calculations.] [3]
- (b) How big a telescope (in terms of diameter) is enough to distinguish between Sirius A and B? [2]

Noise in a Resonator

Consider a circuit comprising an inductor of inductance L and a capacitor with capacitance C . Now an ideal voltmeter is attached with the circuit in parallel with both the elements mentioned above.

- (a) Draw the circuit diagram using the standard symbols embedded in the circuit diagrams. [1]

- (b) If the current in the inductor at any time is I and the charge stored in the capacitor is Q , what is the energy stored in the circuit? [2]

- (c) Typically if the circuit has no EMF source and the system is kept at absolute zero temperature, there won't be any reading on the voltmeter. But the game changes if the system is kept at a nonzero temperature and voltmeter can show nonzero readings. Find the RMS value of the reading on the Voltmeter. Assume that the inductance and capacitance remains unaffected by the temperature. [3]

Thickness by Radiation

An experimentalist measures the thickness of an aluminium plate placed perpendicular to an X-ray beam. Using an X-ray detector and source, measurements are taken at various points right behind the plate after the same duration of X-ray exposure. The number of counts, N , obtained in 6 trials were 1.0×10^4 , 1.02×10^4 , 1.04×10^4 , 1.06×10^4 , 1.08×10^4 , 1.1×10^4 .

(a) What is the standard deviation associated with these measurements?

[2]

(b) Given that

$$N = N_0 e^{-ut}$$

where t is the thickness of the plate, how is a fractional uncertainty in N related to a fractional uncertainty in t ?

[2]